

INTELLIGENT HEALTHCARE DIAGNOSTIC ASSISTANT

Introduction to Artificial Intelligence — Capstone Project

Project	Intelligent Healthcare Diagnostic Assistant
AI Techniques	Intelligent Agents, Logic, Bayesian Diagnosis, ML, DNN, Fuzzy Logic, AI Planning
System Modules	Modules 1–7
Full-System Test	5 patients
Model Evaluation	8 test patients

Important: This is an academic AI prototype and is not a substitute for professional medical diagnosis, triage, or treatment.

1. Executive Summary

The Intelligent Healthcare Diagnostic Assistant is a multi-technique artificial intelligence capstone system designed to demonstrate how different AI approaches can work together in a healthcare diagnostic workflow. The system combines an intelligent agent with a knowledge base, Bayesian diagnostics, a machine-learning classifier, a deep neural network, fuzzy severity assessment, and STRIPS-style treatment planning.

The complete system was integrated through `app.py` and successfully tested with five patient cases. A separate evaluation pipeline evaluated the two learned diagnostic models using eight test patients and generated classification metrics, confusion matrices, a model comparison chart, and a CSV results file.

2. Project Objectives

The project demonstrates intelligent-agent operation, medical knowledge representation, Bayesian reasoning, machine-learning classification, deep learning, fuzzy severity assessment, and AI planning. It also demonstrates how these techniques can be integrated into a single end-to-end application.

3. System Architecture

Module 1 coordinates the system using a Perceive → Think → Act cycle. Modules 2–7 provide specialized knowledge, diagnostic, severity, and planning capabilities. Patient information is represented through the `PatientPercept` structure and is passed through the registered AI modules by the agent.

4. AI Techniques Implemented

Module 1: Intelligent Agent and PEAS framework. Module 2: Medical Knowledge Base. Module 3: Bayesian Diagnostics. Module 4: Machine Learning Diagnostic Classifier. Module 5: Deep Neural Network Diagnostic Model. Module 6: Fuzzy Logic Severity Assessor using temperature, heart rate, and symptom count. Module 7: STRIPS-based treatment planning using breadth-first search.

5. Full-System Integration

The modules were instantiated and registered with `HealthcareDiagnosticAgent` in `app.py`. The agent perceives a patient, runs the registered diagnostic modules, aggregates available diagnostic information, assesses urgency, generates system-level recommendations, and produces a structured report. The integrated application completed a five-patient test successfully.

6. Evaluation Methodology

The evaluation focused on Module 4 and Module 5 because these are learned diagnostic models for which standard classification metrics apply. The evaluator calculates accuracy, weighted precision, weighted recall, weighted F1-score, confusion matrices, and ROC-AUC when probability predictions are supplied.

8. Confusion Matrices and Visualizations

The evaluation pipeline successfully generated two confusion matrices, one model-comparison chart, and `evaluation_results.csv`. These artifacts are stored in the `evaluation/results` directory of the project.

9. Discussion

The results demonstrate that symbolic, probabilistic, machine-learning, fuzzy, and planning approaches can be combined within one modular AI application. Module 4 achieved strong but imperfect performance, while Module 5 achieved perfect scores on the current evaluation set. The fuzzy controller provides gradual severity reasoning rather than simply selecting a diagnosis, while the planner converts diagnosis and urgency information into an action

sequence.

10. Limitations

The system is an academic prototype and is not clinically validated. The learned models were evaluated on the project's available test data. The 100% neural-network score should not be interpreted as guaranteed real-world accuracy. ROC-AUC was not reported because probability predictions were not supplied to the current evaluation call. Treatment plans are demonstration-level AI planning outputs and do not replace clinician-authored protocols.

11. Conclusion

The capstone successfully demonstrates an integrated intelligent healthcare diagnostic assistant using seven AI modules. The application completed its end-to-end five-patient integration test, and the evaluation pipeline generated the required metrics and visualization artifacts for the learned diagnostic models.

7. Evaluation Results

The evaluation contained 8 test patients. The recorded results were:

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Module 4 — ML Classifier	87.50%	81.25%	87.50%	83.33%	N/A
Module 5 — Neural Network	100.00%	100.00%	100.00%	100.00%	N/A

The neural-network result is performance on the available evaluation data, not evidence of perfect real-world clinical accuracy. ROC-AUC is N/A because probability-score arrays were not supplied to the evaluator.

12. Final Deliverables Checklist

- ✓ All 7 AI modules implemented
- ✓ Modules integrated through app.py
- ✓ Full-system test completed on 5 patients
- ✓ Evaluation metrics generated
- ✓ Confusion matrices generated
- ✓ Model comparison chart generated
- ✓ evaluation_results.csv generated
- ✓ Final report generated
- README.md update
- 10-minute live demonstration preparation